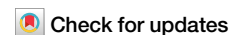


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Peat wildfires during the Early Cretaceous Aptian–Albian of the Erlian Basin in Inner Mongolia, China

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Parts of the Cretaceous period were characterized by widespread wildfire activity. Wildfire evidence can serve as a proxy for paleoclimatic conditions. Charcoal and pyrolyzed polycyclic aromatic hydrocarbons preserved in sedimentary records are widely recognized as reliable proxies for wildfire activity, offering insights into the nature of wildfires. This study employed a multi-method approach, integrating polycyclic aromatic hydrocarbons analysis, temperature estimation equations for peat combustion, and microscopic images. This methodology was applied to the No. 6 coal from the Shengli Coalfield, Inner Mongolia, China. Results indicate that wildfire temperatures during peat accumulation in the Erlian Basin ranged from approximately 300 to 600 °C. Carbonized fossil assemblages from the Aptian–Albian are globally dominated by ferns and gymnosperms, with angiosperms occurring only rarely.

Fire is a fundamental Earth system process, intricately linked to vegetation dynamics and climate variability^{1–3}. The earliest documented wildfires date back to the Silurian period^{4–6}, and have been recorded in every subsequent period of geologic history⁷. Wildfires are thought to have been widespread throughout much of the Cretaceous period^{8–11}. Cretaceous wildfires have been associated with a number of factors, including high atmospheric oxygen levels, low precipitation climatic conditions, and changes in vegetation composition^{7,11}. The Aptian–Albian interval of the Cretaceous was a crucial period in the evolution of angiosperms¹². Several studies suggest that a prevalence of wildfires during the Cretaceous may have contributed to the rapid diversification of angiosperms^{2,13–15}. Understanding the distribution of Aptian–Albian wildfire events, the types of plant groups affected, and the climatic conditions of the time will enhance our capacity to predict and mitigate wildfires driven by contemporary climate anomalies.

Wildfires produce various combustion materials, including charcoal, smoke, and pyrolyzed polycyclic aromatic hydrocarbons (PAHs)^{16,17}. Charcoal is a product of incomplete combustion, which can be preserved in sediments^{8,11,16,18,19}. In coal, layers, lenses, or bands of fossilized charcoal can easily range from 0.5 to over 3 cm thick and are termed fusain^{18,20,21}. Fusain in coal is considered an effective geological indicator of wildfire activity^{22–24}. PAHs can be categorized into three primary groups based on their origin: petroleum origin, mixed origin, and pyrolytic origin^{25–29}. Petroleum PAHs are primarily associated with oil deposits, while pyrolytic PAHs are formed during combustion processes and typically consisting of 3-, 4-, and 5-ring

structures with strong antioxidant properties^{30,31}. Due to their chemical stability, pyrolytic PAHs are considered reliable biomarkers for forest or coal fires^{31,32}. Thus, paleo-wildfire information can be inferred by examining charcoal and pyrolyzed PAHs in sediments, including coal, and can provide insight into paleoclimate conditions^{22,23,26,29}.

The Shengli Formation in the Erlian Basin was deposited during the Aptian–Albian stages of the Early Cretaceous^{33,34}. Previous studies have found that the No. 6 coal contains high concentrations of fusinites and semifusinites^{34,35}. This study integrates coal petrology, geochemical data, and Gas Chromatography–Mass Spectrometry (GC–MS) analysis to elucidate the evolution of wildfire activity during the deposition of the No. 6 coal at two locations within the Erlian Basin (Supplementary Fig. 1). Analysis of petrographic and geochemical indicators from coal samples at two locations in the Shengli Coalfield enable inference of the wildfire temperature range during the Aptian–Albian period in the Erlian Basin.

Results

Maceral composition

Inertinite is the dominant maceral group in coals from both locations in the Erlian Basin, with fusinite and semifusinite as the primary components. In coals from Location #1, inertinite content ranges from 6% to 66%, with a mean value of 53%. In contrast, coals from Location #2 exhibit a broader range of inertinite content (from 3% to 70%), but a lower mean value of 41% (maceral compositions data from Dai et al.^{34,35}).

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Inertinite reflectance values and calculated combustion temperatures

Inertinite reflectance values for coal samples from Location #1 range from 0.93% to 2.79%, with an average of 1.63%, while those from Location #2 range from 0.45% to 4.48%, with an average of 1.38% (Supplementary Data 1 Table 1). Box plots for Locations #1 and #2, arranged in stratigraphic order, illustrate the statistical distribution of inertinite reflectance values (Supplementary Fig. 2). Although only a portion of the profile samples were collected at Location #1, the inertinite reflectance in these samples is slightly higher than that observed in most samples from Location #2. A simple Student's *t*-test confirms that the inertinite reflectances of coals from the two locations differ significantly. All inertinite reflectance data and their normal distribution curves are given in Supplementary Data 2 and Supplementary Fig. 3 (see Supplemental Data).

The distribution of combustion temperatures for each sample, calculated using 11 equations (Supplementary Data 1 Table 2) is shown as stratigraphically arranged box plots (Fig. 1; see Supplementary Data 2). Averages by location for each of the 11 temperature formulas show a wide range (Supplementary Data 1 Table 3) from 337 to 599 °C (the temperature difference is $\Delta T = 262$ °C) for Location #1 ($n = 14,080$ temperature estimates) and from 316 to 523 °C ($\Delta T = 207$ °C) for Location #2 ($n = 35,618$ temperature estimates) (Supplementary Data 1 Table 3). At both locations, the lowest predicted temperature is derived from the Guo and Bustin³⁶ formula for decayed wood, while the highest predicted temperature is given by the formula developed by McParland et al.³⁷. Predicted temperatures exceeding 2000 °C (only 5 instances) are

considered improbable (for reasons discussed later) and have been excluded from graphical representations, but they are included in the Supplemental Data.

The average predicted temperature of formation for Locations #1 and #2 is 450 °C and 418 °C, respectively (Supplementary Data 1 Table 4). A Student's *t*-test reveals a statistically significant difference between the two populations, indicating a phase of slightly higher combustion temperatures in the paleomire at Location #1.

Polycyclic aromatic hydrocarbons

This GC-MS analysis focuses on detecting aromatic compounds with 3 to 6 rings and quantifying their absolute concentration (Supplementary Data 1 Table 5). In samples from both locations, the concentrations of phenanthrene (P), retene (Ret), fluoranthene (Fla), pyrene (Pyr), chrysene (Ch), benzo[b,k]fluoranthene (BFla), benzo[e]pyrene (BeP), benzo[a]pyrene (BaP), and perylene (Pery) were relatively higher than those of dibenzothiophene (DBT), dibenzofuran (DBF), benzo[g,h,i]perylene (BPer), and indeno[1,2,3-c,d]pyrene (InPyr). Additionally, benzo[a]anthracene (BaAn) concentrations are higher in samples from Location #1 than in those from Location #2. However, coronene was not detected in any samples from either location.

Discussion

The relatively high concentrations of Fla, Pyr, BFla, BaP, and BeP in samples from both locations indicate that persistent wildfire activity occurred in the peat during the formation of the No. 6 coal seam. However, the absence of

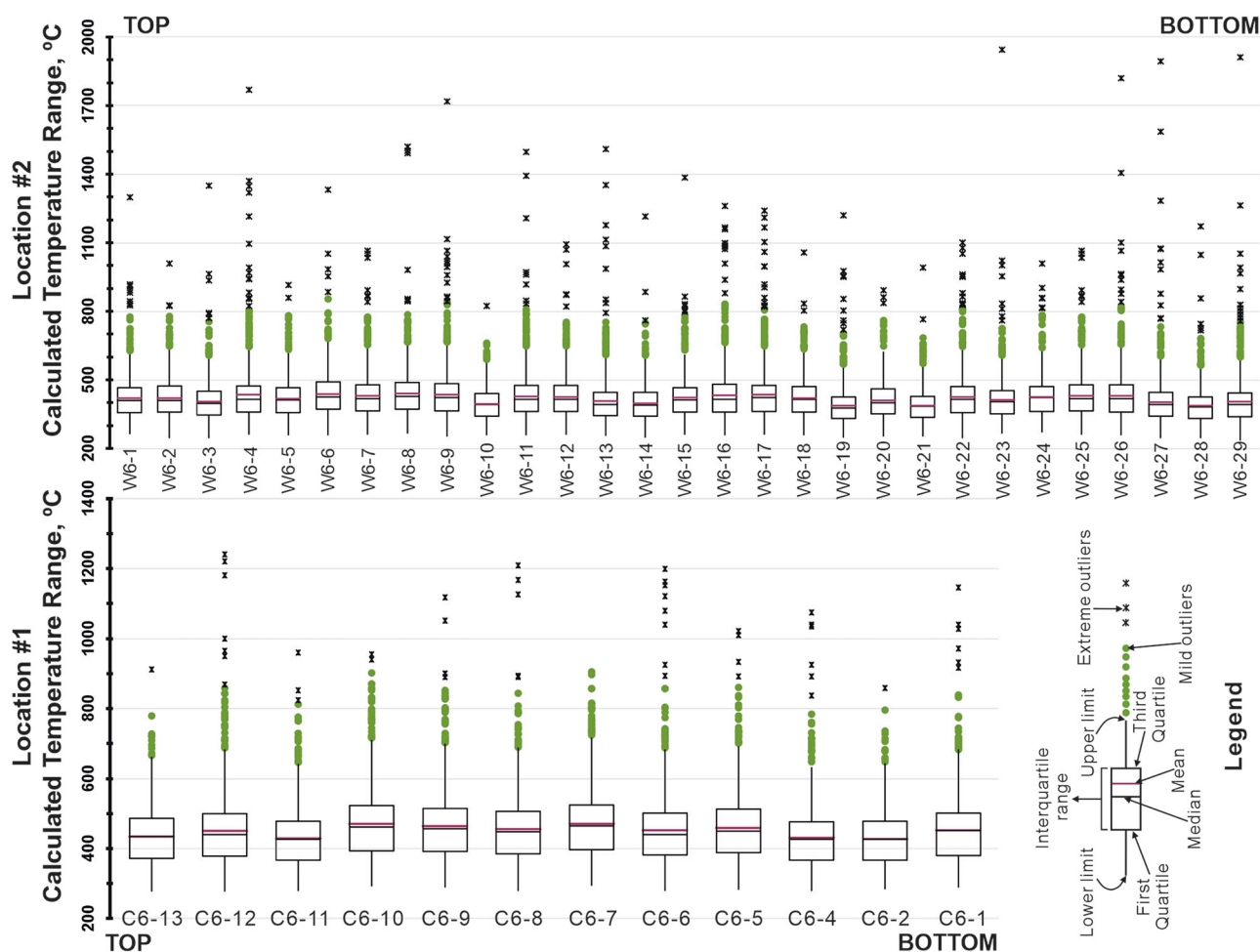


Fig. 1 | Box plots of wildfire temperatures derived from inertinite reflectance for Locations #1 and #2. In boxplot, the box represents the interquartile range (IQR), or the middle 50% of the data. Crosses and green circles outside the box indicate

outliers. Median, the value located at the middle position after sorting the data from smallest to largest (or from largest to smallest).

coronene, an indicator of high-temperature combustion products³⁸, indicates that wildfire temperatures in the area were not excessively high.

To further elucidate the sources of PAHs in the No. 6 coal from the Shengli Coalfield, four diagnostic ratios were employed: InPyr/(Bper + InPyr), Fla/(Pyr + Fla), Fla/Pyr, and MP/P (Supplementary Data 1 Table 6). Analysis of PAHs ratios from samples at both locations indicates that most of the InPyr, Bper, Fla, Pyr, and P are of pyrolytic origins (Fig. 2). However, some of the Fla and Pyr (e.g., C6-3, C6-12), and P (e.g., W6-11, W6-18) show mixed origins. Only the P in sample W6-9 was derived from petroleum sources, likely due to transformation of P during diagenesis (Fig. 2). Biodegradable compounds such as DBT and DBF, which are derived from land plant polysaccharides and lignin, were also detected in addition to the previously mentioned PAHs that indicate combustion events^{32,39,40}. The presence of PAHs, DBT, and DBF provides further evidence of wildfire activity at both sampling locations, and most likely throughout the Shengli Coalfield, during the Aptian–Albian period.

Charcoal, which results from incomplete combustion during wildfires, has been extensively employed by many researchers to investigate the wildfire impacts on palaeoenvironments^{22,41}. The origin of fusinite and semifusinite has sparked debate among researchers. Fusinite, which is macroscopically identifiable layers of charcoal, is generally composed of inertinite macerals (predominantly fusinite and semifusinite), but it may also contain vitrinite (or huminite) and liptinite in varying proportions^{42,43}. Some researchers have proposed that fusinite and semifusinite, in particular, are by-products of wildfire^{18,22,44}. Conversely, others have suggested that fusinite may have formed through the decomposition, oxidation, and dehydration of plant tissue, facilitated by the activity of fungi and bacteria, ultimately leading to the formation of degradofusinite^{45–48}. In addition, Moroeng et al.⁴² suggested that some inertinites may have been altered after combustion by microbial activity, in much the same way that modern charcoal has been found to be altered^{49–51}.

In samples from both locations studied, inertinite macerals are primarily composed of fusinite and semifusinite, many of which exhibit typical pyrofusinite traits such as intact cellular structure, faint yellow color, and high reflectance⁵² (Fig. 3). Additionally, Dai et al.³⁵ observed that even some macrinites had the same reflectance as the surrounding semifusinite in samples from Location #2, indicating that they underwent the same charring process as the surrounding semifusinite.

The presence of pyrofusinite in the samples from both locations in the Shengli coalfield supports the PAH results, indicating that wildfires were common and likely frequent phenomena. These wildfires led to inertinite content being of significant proportions in the paleomires.

Wildfires can be categorized into three main types based on where they are actually combusting: ground (or smoldering), surface, and crown (or canopy) fires^{53–58}. Ground fires mainly consume organic material beneath the surface litter, spreading slowly and lasting for extended periods. These fires typically maintain a combustion temperature below 600 °C^{53,59}. Surface fires, on the other hand, ignite forest litter or low shrubs and herbs, spreading rapidly and lasting for a considerable duration. The temperature of surface fires generally ranges from 500 to 800 °C^{53,59,60}, although some have been reported to exceed 1000 °C⁶¹. Crown fires are wildfires that spread through the tree canopy. The primary fuel for crown fires consists of the canopy of vegetation, shrubs, and forest plants located more than 1 m above the ground. Crown fires are extremely destructive, with temperatures typically ranging from 500 to 800 °C^{60,62} and occasionally exceeding 1330 °C⁵⁶. These fires can be caused by surface fires that ignite plants wrapped around tree trunks^{54,60,63}.

Carbonization experiments have demonstrated that inertinite reflectance increases as combustion temperature rises^{18,64}. This relationship can be leveraged to infer the burning temperature of ancient wildfires. Although there is not a perfect linear relationship between inertinite reflectance and combustion temperature, this relationship has been approximated using regression equations. One such equation was developed by Jones⁶⁴ and has subsequently been applied by other researchers, including Wang et al.²² and Zhao et al.²⁶.

However, as noted by Moroeng et al.⁴² several laboratory combustion experiments have compared the reflectance of fusinitized plant remains at different temperatures. Each of these studies has slightly different correlations between temperature and fusinite reflectance. Using all of the studies allows the uncertainty in the temperature of formation of the inertinites to be estimated through statistical analysis of the mean and standard deviation (Supplementary Data 1 Table 3; Fig. 1).

Although both Locations #1 and #2 have significantly high temperature outliers, the means and upper and lower quartiles are surprisingly consistent between samples collected at both sites. It is suggested that fire type (i.e., ground, surface, and crown) be primarily based on the temperature range indicated by the upper and lower quartiles. The outliers (mild and extreme) do not represent a volumetrically significant proportion, and it seems unlikely that they alone could indicate a different fire type. Most of the outliers in this study are likely representative of temperature hotspots that can occur in all fires. However, these hotspots are localized and not diagnostic for any particular fire type. It is recognized that the distribution of temperature in fires is complex. A process called flaming can occur in smoldering peat fires, reaching temperatures in excess of 700 °C⁶⁵.

Temperatures observed in modern fire overlap⁴², and discrimination between types based on temperature is generally considered unreliable⁶². However, given that the average calculated temperatures at both locations fall within the low to mid 400 °C, it seems reasonable to infer that the paleomires were primarily affected by ground fires. Most studies on modern fires demonstrate that almost all peat fires are ground fires^{63,65–67}.

Palynological evidence from the No. 6 coal seam³⁴ indicates that bryophytes, pteridophytes, and gymnosperms were the main plants present in the paleomires and thus were most likely affected by wildfires, regardless of the type of fire. This is consistent with other studies of Early Cretaceous rift-basin coals in Inner Mongolia, which also contain significant proportions of inertinite and are ecologically similar to vegetation inferred for the No. 6 seam in the Shengli coalfield^{68,69}. This pattern holds global relevance, as these plant groups were dominant components of terrestrial ecosystems during the Early Cretaceous^{70–74}.

Plant cell walls consist of primary walls, secondary walls, and a middle lamella, all of which undergo morphological changes during charring¹⁶. At lower combustion temperatures, the middle lamella remains clearly visible within the cell walls. With increasing temperature, however, these lamellar structures gradually disappear, resulting in cell walls with a more homogeneous texture. The homogenization of cell walls depends on the plant species but typically occurs between 280 °C and 320 °C¹⁶. When temperatures exceed 370 °C, plant cell walls begin to rupture⁷⁵; at ~600 °C, the cell wall breaks into fragments known as inertodetrinite when preserved in coal¹⁶. Scanning electron microscope (SEM) imagery shows the homogenization of fusinite cell walls in samples from both locations in this study (Fig. 3). Based only on cell wall morphology, the fire temperatures are interpreted to have been relatively low. This interpretation is consistent with the generally low temperature nature of fires affecting the paleomires at Locations #1 and #2, supporting the conclusion that these were primarily ground or smoldering fires.

The No. 6 coal in the Shengli Coalfield was deposited during the Aptian–Albian period^{33,34}, a crucial epoch when wildfires were common globally and which also coincided with evolutionary changes in gymnosperms and the early diversification of angiosperms^{8,11}. To better understand Early Cretaceous wildfires during the Aptian–Albian period and to provide insights from studies across both the Northern and Southern hemispheres, we collected and analyzed data on wildfire occurrences (Supplementary Table 1). The majority of reported wildfires from the Aptian–Albian period are concentrated in northeastern China, Inner Mongolia, North America, the Atlantic coast of Europe, Australia, and India (Supplementary Table 1). It is important to note that this does not imply these were the only regions where wildfires occurred during this period. Absence of wildfire records may be attributed to several factors, including low atmospheric oxygen concentration, sedimentary conditions unsuitable for charcoal preservation, and insufficient fuel availability, among other

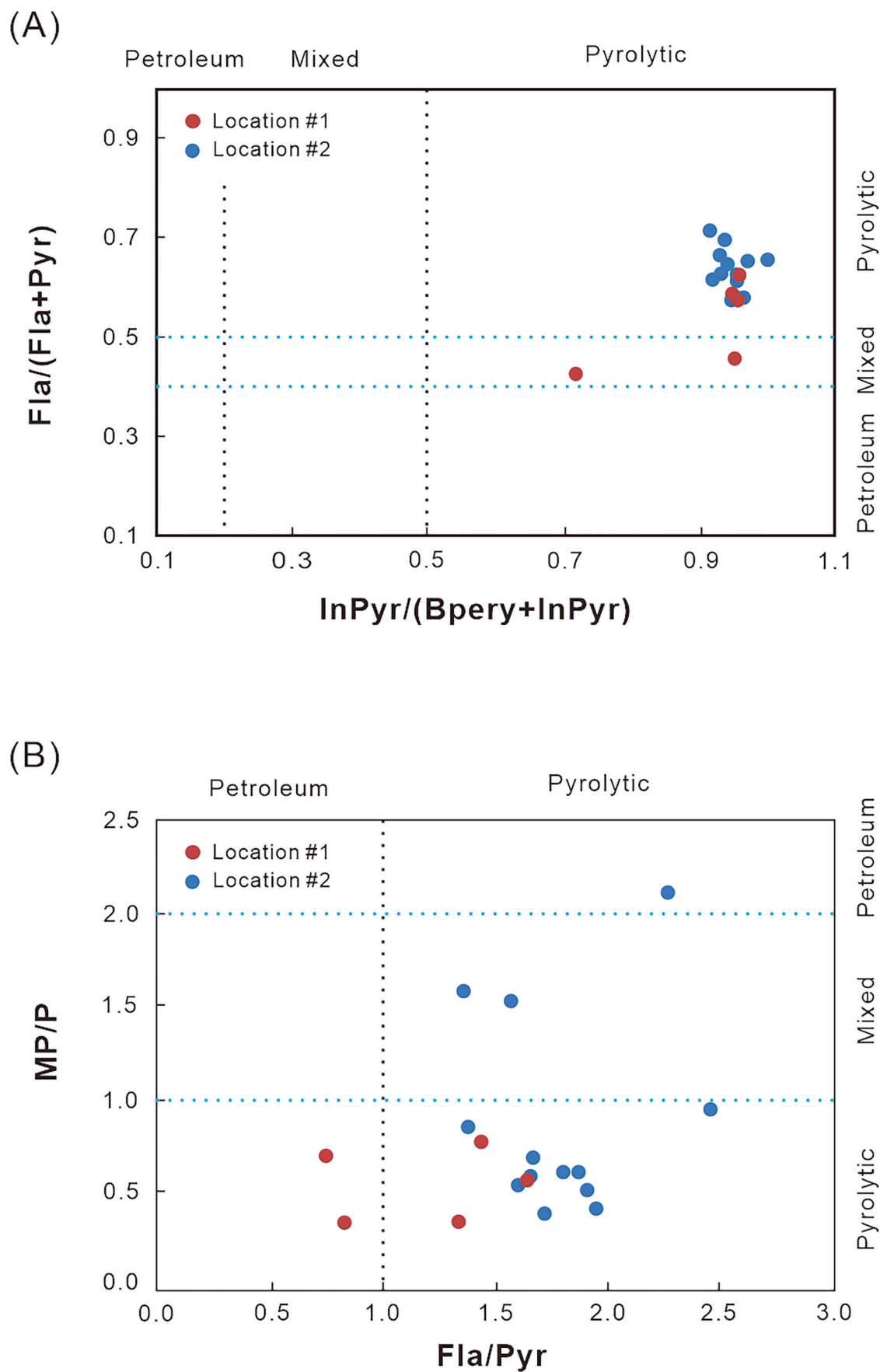
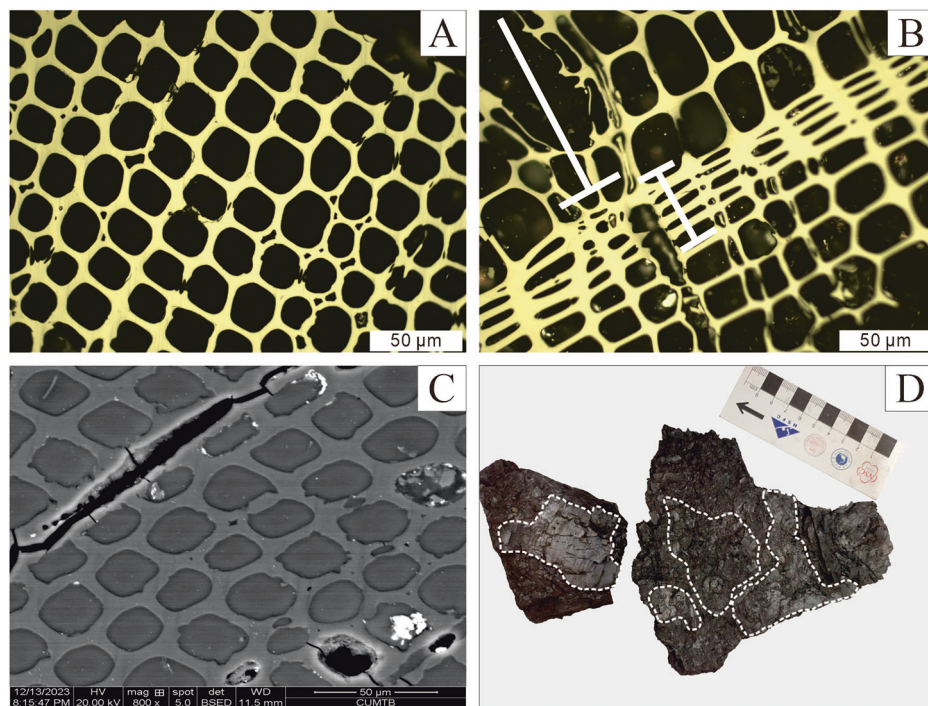


Fig. 2 | Diagnostic plots indicating the origin of PAHs. A The plots of $\text{InPyr}/(\text{Bperyl} + \text{InPyr})$ vs. $\text{Fla}/(\text{Fla} + \text{Pyr})$. **B** The plots of Fla/Pyr vs. MP/P . Fla fluoranthene, Pyr pyrene, P phenanthrene, MP methylphenanthrene, $\text{MP} = 3\text{-MP} + 2\text{-MP} + 9\text{-MP} + 1\text{-MP}$; BPeryl benzo[g,h,i]perylene; InPyr indeno[c,d]pyrene.

Fig. 3 | Fusinite and charcoallified wood fragment in the Shengli Coalfield. **A** Fusinite in sample W6-7. Reflected light, oil immersion. **B** Fusinite with growth rings (white lines); sample W6-13. Reflected light, oil immersion. **C** SEM back-scattered electron images of fusinite in sample W6-7. **D** A photo of charcoallified wood fragment (The area enclosed by the white dashed line).



possible factors^{11,76}. However, in China, records of wildfires during the Aptian–Albian period have been reported almost exclusively in Inner Mongolia, Northeast China, and Gansu province.

A binary plot (Fig. 4) of wildfire temperatures and inertinite contents reported from Inner Mongolia and Northeast China shows that inertinite contents range from 0.2 to 85.0 vol%. Inertinite reflectance data indicate that the wildfire temperatures, as inferred from the mean of the 11 equations, likely ranged from ~300 to 630 °C, with most points falling between 300 and 500 °C. This indicates that the Early Cretaceous Aptian–Albian wildfires in Inner Mongolia and Northeast China were not particularly intense and may have been predominantly ground (i.e., smoldering) fires. No inertinite reflectance data from coal samples outside of China were identified during this data collection. However, the well-preserved charred plants and homogenized cell walls of other areas¹⁷⁷ may indicate that wildfire temperatures were ~300 °C and did not exceed ~600 °C^{16,75}. Alternatively, it is possible that fires exceeding 600 °C do not produce sufficient quantities of well-preserved charcoal suitable for analysis, although few quantitative studies support this viewpoint¹⁶.

The origin of angiosperms remains a topic of debate⁷⁸. In China and especially northeast and northern China, angiosperms are well documented in the Early Cretaceous^{79–84}. Gravendyck et al.⁷⁸ and Kvaček et al.⁸⁵ have proposed a northern Gondwanan origin, possibly in the Iberian Peninsula, such as Portugal. They suggested a gradual evolutionary expansion from low paleolatitudes to mid-latitudes, eventually reaching high-latitude regions such as the Arctic. However, most Aptian–Albian charcoal found in coal is identified as gymnosperms or as unidentified material, possibly ferns⁶⁹. This implies that although angiosperms may have originated locally, they had not yet become widespread within peat swamp ecosystems or existed only in minor proportions during the Aptian–Albian, resulting in limited preservation in the fossil record. Interestingly, fossil charcoal attributable to angiosperms has been identified in the United States of America, Italy, and Portugal^{86–88}, which supports other interdisciplinary studies that have established angiosperms had already achieved a broad intercontinental distribution by the Early Cretaceous⁸⁹.

Bond and Scott² have proposed previously that abundant wildfires during the Cretaceous may have contributed to the evolution of angiosperms. Although an interesting hypothesis, supporting data is still needed.

In the study area, growth rings (Fig. 3B) indicate seasonal variation in temperature and/or precipitation, and other studies^{22,24,34,35,42,46,69,90} indicate that Early Cretaceous coal beds in Inner Mongolia, as well as the study area sites, have been affected by wildfire. Although this study does not address paleoflora changes, the objective is to document the frequency of fire and fire temperature in the studied coals (i.e., paleomires). The significance of this study is that it adds to the growing database of fire processes during the Early Cretaceous.

Methods

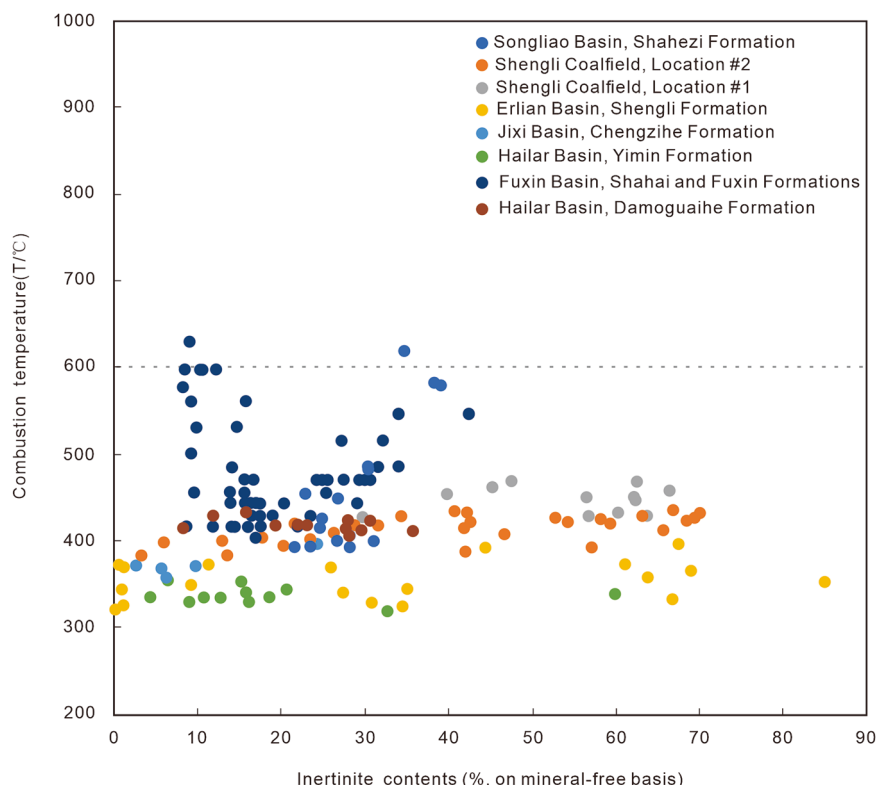
Samples

Coal samples for this study were collected from two locations within the Shengli Coalfield (Supplementary Fig. 1C). Location #1 is in the south-western part of the coalfield and was previously described by Dai et al.³⁴. A total of 13 coal samples were collected from Location #1, all from No. 6 coal seam, and numbered sequentially from bottom to top as C6-1 to C6-13. Notably, two partings are present (one between samples C6-6 and C6-7, and another between C6-12 and C6-13) with thicknesses of 10 cm and 3 cm, respectively. Location #2, situated just northeast of sample Location #1 in the area known as the No. 1 Pit, was previously described by Dai et al.³⁵. At this location, 32 samples were collected from No. 6 coal seam, comprising 29 coal bench samples, one carbonaceous mudstone parting (W6-30), and mudstone samples from the seam roof (W6-R) and floor (W6-F). Coal samples from Location #2 are sequentially numbered from top to bottom as W6-1 to W6-29. Only coal samples were analyzed in this study; parting, roof, and floor samples were excluded. All samples were collected from freshly exposed faces in accordance with the Chinese National Standard GB/T 482–2008⁹¹ to ensure consistency and quality. Unlike ISO 14180:2023⁹², which requires the exposed width of the coal seam to be at least twice the intended sampling width, GB/T 482–2008⁹¹ mandates that for coal seams thicker than 1.3 m, samples should be taken at a width of 0.10 m and a depth of 0.05 m. After sampling, all materials were promptly sealed and stored to prevent oxidation and contamination.

Inertinite reflectance and scanning electron microscope

According to the GB/T 16773–2008⁹³ standard, each coal sample was crushed into particles smaller than 1 mm and subsequently embedded in epoxy

Fig. 4 | The inertinite contents and combustion temperature reported from Inner Mongolia and Northeast China. The temperature is the average of the results calculated from 11 equations. Inertinite contents and reflectance data from Supplementary Table 1 and this study.



resin to make polished blocks for electron and optical microscopic analysis. These blocks were examined under both reflected and fluorescent light, utilizing a Leica (DM4P) microscope outfitted with a Craic microspectrophotometer and 50× oil immersion objectives. Previous analyses conducted by Dai et al.^{34,35} encompassed microscopic composition analysis and maximum huminite reflectance measurements of the samples studied. Hence, only inertinite reflectance measurements were conducted in this study. To ensure accuracy, the microscope was calibrated against YAG (0.906 %R₀), GGG (1.722 %R₀), and optical black (zero) standards at room temperature. For the inertinite reflectance measurements, at least 100 effective points were utilized within each block, except for sample W6-15, where only 99 test points were completed due to the low inertinite content. The mechanical stage moves the block in fixed steps according to the pre-determined number of measurement points. The reflectance data from all effective points within each polished block are then averaged to obtain the final reflectance values. Compared to carbonized plant fossils preserved in other sedimentary rocks, coal's fragile nature limits the preservation of complete plant tissues. Therefore, under the microscope, it is difficult to identify specific plant taxa or to determine which parts of the plant have been preserved. Additionally, certain polished blocks were examined using a field emission scanning electron microscope equipped with energy-dispersive X-ray spectrometry (SEM-EDS; FEI Quanta™ 650 FEG) to assess the homogenization of cell walls. For the SEM-EDS test, blocks were carbon-coated using a Quorum Q150T ES sputtering coater and then mounted on standard aluminum SEM stubs using conductive carbon tape. The beam voltage was set to 20.0 kV, with an aperture of 6 and a spot size ranging from 4.5 to 5.0 nm. Images were captured using a retractable solid-state back-scattered electron detector.

Polycyclic aromatic hydrocarbons (PAHs) determination

To extract soluble organic matter, 10 grams of the sample (200 mesh) was transferred to an extraction tank. Next, 20–50 mg of the soluble organic matter was thoroughly mixed with 30 ml of *n*-hexane. The mixture was allowed to stand for more than 12 h to allow asphaltenes to precipitate. Following asphaltene removal, saturated hydrocarbons, aromatic

hydrocarbons, and nonhydrocarbons were eluted using *n*-hexane, a 2:1 mixture of dichloromethane to *n*-hexane, and dichloromethane, respectively. Subsequently, the filtrate was evaporated at temperatures not exceeding 40 °C until near dryness. Aromatic hydrocarbons were analyzed using an Agilent 7890BGC-5977MS instrument, with D8 dibenzothiophene (D8-DBT) as the internal standard. During testing, the injection port temperature was maintained at 300 °C. Helium (99.999%) was used as the carrier gas at a flow rate of 1.0 mL/min. The initial detector temperature was set at 80 °C for 1 min, followed by a ramp to 310 °C at a rate of 3 °C/min, and then held isothermally at 310 °C for 16 min.

Wildfire temperature estimation

Estimation of wildfire temperature from inertinite reflectance is not well constrained. In the limited laboratory experiments available, it has been shown that different plant types and different parts of the same plants can result in varying charcoal (i.e., inertinite) reflectance at the same combustion temperature⁹⁴. Since it is extremely difficult, if not impossible, to determine the plant parts and types from which inertinite in coal is derived, six studies were used to estimate wildfire temperatures from inertinite reflectance (Supplementary Data 1 Table 2). Commonly, only the relationship established by Jones⁶⁴ is used in many published articles, presumably because that paper provides a numeric formula. In this study, numerical relationships were derived from the published graphical results of temperature versus charcoal reflectance reported in five additional studies. In the study by McParland et al.³⁷, they documented the relationship between temperature and charcoal reflectance from two different plant types. For one of the plant types, they used both the xylem and cortical parts in their burn experiments. In the combustion experiments by Guo and Bustin³⁶, they derived relationships between temperature and charcoal reflectance for both undecayed and decayed wood. Finally, in the studies by Ascough et al.⁹⁵ and Hudspith et al.⁹⁴, both groups conducted multiple experiments that resulted in a range of charcoal reflectance values for plant material at the same combustion temperature. For these two studies, we derived formulas for both low reflectance and high reflectance scenarios. Thus, using the six studies, each reflectance measurement in this study received 11 estimates for its possible

temperature of formation based on the formulas provided in Supplementary Data 1 Table 2.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in this study can be found in the Supplementary Information. The Supplementary datasets can be obtained from <https://zenodo.org/records/17393938>.

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Competing interests

The authors declare no competing interests.

Additional information

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